

BOARDING HOUSE MANAGEMENT INFORMATION SYSTEM
DATABASE FINAL EXAM REPORT



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D-4 INFORMATION MANAGEMENT STUDY PROGRAM

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A. Selected Topic

- e. Rooming House Management Information System (Sistem Informasi Pengelolaan Kos)

B. Business Processes and Modules

1. Business Processes:

- a. **Room Booking & Registration:** Prospective tenants browse available rooms (checking numbers, types, and pricing). If interested, they register by filling out personal details, selecting a room, and completing the initial payment or first-month rent.
- b. **Tenant & Room Management:** The Admin/Owner validates the payment, updates the room status to "Occupied", and actively maintains tenant records.
- c. **Monthly Rent Payment:** Every month, the system or admin generates a rental invoice. Tenants pay their rent, upload a transfer receipt, and the admin verifies the transaction to update the status to "Paid".
- d. **Maintenance & Complaint Management:** Tenants can lodge complaints regarding facility issues (e.g., broken AC). The owner tracks these complaints and records operational maintenance expenses.

2. System Modules:

- a. **User & Authentication Module:** Manages login credentials and access levels for both Admins/Owners and Tenants.
- b. **Room Management Module:** Manages master room data (room numbers, room types, monthly rates, and availability status).
- c. **Tenant Registration Module:** Manages active tenant data, check-in dates, and historical records.
- d. **Payment Transaction Module:** Manages monthly invoice generation, rental payments, and transaction history.
- e. **Complaint & Maintenance Module:** Tracks facility damage reports submitted by tenants.

3. Involved Parties in Each Module

1. Owner / Admin:

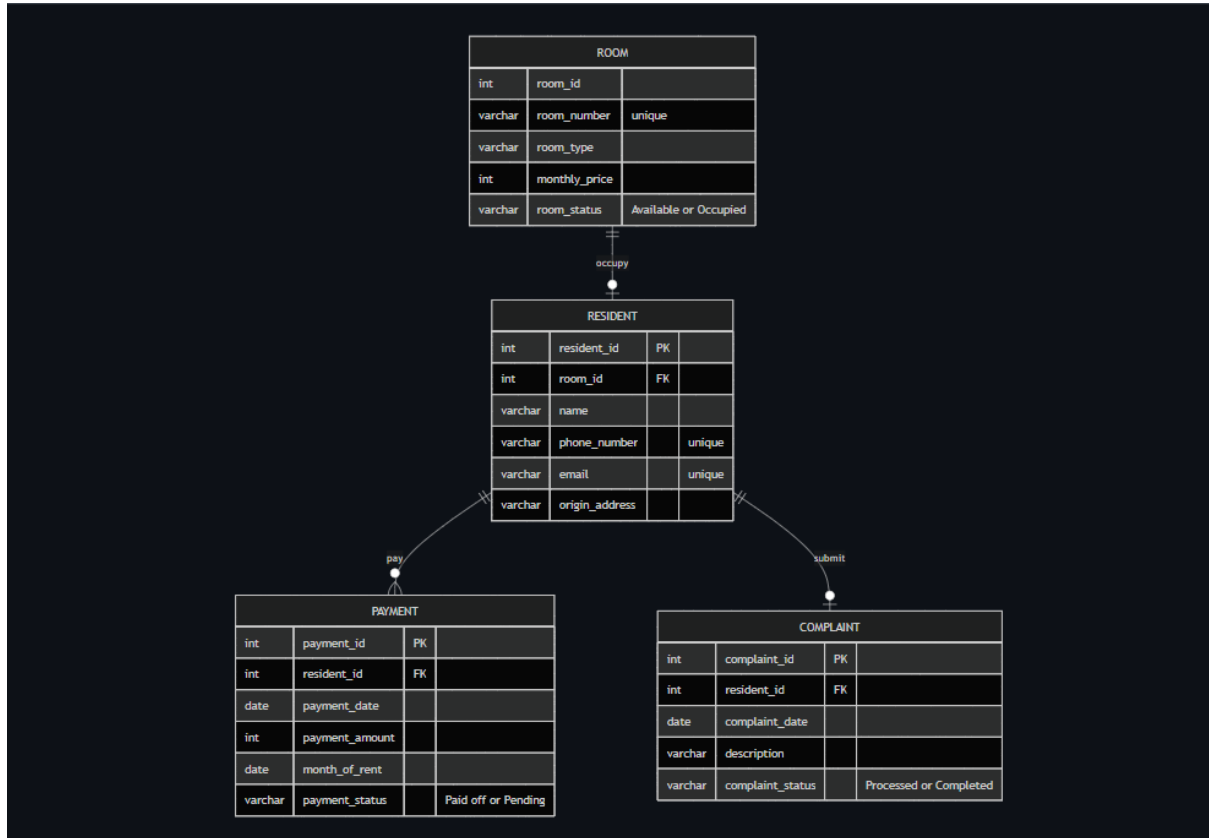
- Involved in all modules.
- Responsibilities: Managing room records, verifying rental payments, updating room status, and tracking maintenance reports.

2. Tenant (Resident):

- Involved in the User, Registration, Transaction, and Complaint modules.

- Responsibilities: Providing personal information, selecting a room, paying monthly bills, viewing payment history, and reporting broken facilities.

4 & 5. Entity Relationship Diagram (ERD) & Cardinality



A. Relationships & Cardinality Details:

1. Room to Resident (1 to 1):

- *Explanation:* One room can only be occupied by one active resident at a time, and one resident can only rent one room at a time.
- *Cardinality:* 1 : 1

2. Resident to Payment (1 to Many):

- *Explanation:* One resident can make multiple rental payments over time (monthly), but a single payment transaction record belongs strictly to one resident.
- *Cardinality:* 1 : N

3. Resident to Complaint (1 to Many):

- *Explanation:* One resident can submit multiple complaints over time, while a single complaint record is filed by exactly one resident.
- *Cardinality:* 1 : N

6. Database Normalization Process

A. Unnormalized Form (UNF)

Column Name	Key Type	Description
id_kos_all	Primary Key (PK)	Unique ID for the raw unstructured log row
room_id	-	Unique ID for the room
room_number	-	Room number or room code
room_type	-	Room type/amenities (e.g., AC, Private Bathroom)
monthly_price	-	Monthly rental rate for the room
room_status	-	Current status of the room (Available/Occupied)
resident_id	-	Unique ID for the resident
resident_name	-	Resident's full name
phone_number	-	Resident's active phone number
email	-	Resident's email address
origin_address	-	Resident's original home address
payment_id	-	Unique ID for the payment transaction
payment_date	-	Date when the payment was made
payment_amount	-	Total amount of money paid

Column Name	Key Type	Description
month_of_rent	-	Target month/year for the rent (e.g., "June 2026")
payment_status	-	Transaction status (Pending/Paid)
complaint_id	-	Unique ID for the complaint report
complaint_date	-	Date when the complaint was submitted
complaint_desc	-	Detailed description of the issue or damage
complaint_status	-	Resolution status (In-Progress/Completed)

B. First Normal Form (1NF)

The 1NF table structure remains identical to the UNF table shown above, with the strict enforcement that all attributes contain only atomic (single) values, and no repeating groups exist in any row.

C. Second Normal Form (2NF)

To eliminate partial dependencies, the large table is broken down into core master entities so that non-key attributes depend entirely on the Primary Key.

1. Room Master Table (tb_room)

Column Name	Key Type	Description
room_id	Primary Key (PK)	Unique identifier for each room
room_number	-	Room designation code or number
room_type	-	Room category or specifications
monthly_price	-	Monthly rental fee
room_status	-	Current availability status

2. Resident Master Table (tb_resident)

Column Name	Key Type	Description
resident_id	Primary Key (PK)	Unique identifier for each resident
room_id	Foreign Key (FK)	Reference to the room occupied by the resident
resident_name	-	Full name of the resident
phone_number	-	Active contact number
email	-	Email address
origin_address	-	Home/origin address

D. Third Normal Form (3NF) - Final Schema

To eliminate transitive dependencies, transaction logs (Payments and Complaints) that previously relied transitively on the resident are isolated into their own dedicated operational tables.

1. Room Master Table (tb_room)

Column Name	Key Type	Description
room_id	Primary Key (PK)	Unique identifier for each room
room_number	-	Room designation code or number
room_type	-	Room category or specifications
monthly_price	-	Monthly rental fee
room_status	-	Current availability status

2. Resident Master Table (tb_resident)

Column Name	Key Type	Description
resident_id	Primary Key (PK)	Unique identifier for each resident

Column Name	Key Type	Description
room_id	Foreign Key (FK)	Reference to the room occupied by the resident
resident_name	-	Full name of the resident
phone_number	-	Active contact number
email	-	Email address
origin_address	-	Home/origin address

3. Payment Transaction Table (tb_payment)

Column Name	Key Type	Description
payment_id	Primary Key (PK)	Unique identifier for the transaction
resident_id	Foreign Key (FK)	Reference to the resident who paid
payment_date	-	Date of the transaction
payment_amount	-	Total amount paid
month_of_rent	-	Month and year covered by the payment
payment_status	-	Status of the payment (Pending/Paid)

4. Complaint Management Table (tb_complaint)

Column Name	Key Type	Description
complaint_id	Primary Key (PK)	Unique identifier for the complaint file
resident_id	Foreign Key (FK)	Reference to the resident filing the report
complaint_date	-	Date the complaint was logged

Column Name	Key Type	Description
complaint_desc	-	Full details of the maintenance request
complaint_status	-	Progress status (In-Progress/Completed)

7. Database Implementation (DDL SQL)

SQL

```
CREATE DATABASE db_boarding_house;
```

```
USE db_boarding_house;
```

```
CREATE TABLE tb_room (
    room_id INT AUTO_INCREMENT,
    room_number VARCHAR(10) NOT NULL,
    room_type VARCHAR(50) NOT NULL,
    monthly_price INT NOT NULL,
    room_status ENUM('Available', 'Occupied') DEFAULT 'Available',
    PRIMARY KEY (room_id)
) ENGINE=InnoDB;
```

```
CREATE TABLE tb_resident (
    resident_id INT AUTO_INCREMENT,
    room_id INT UNIQUE,
    resident_name VARCHAR(100) NOT NULL,
    phone_number VARCHAR(15) NOT NULL,
    email VARCHAR(100),
    origin_address TEXT,
    PRIMARY KEY (resident_id),
    FOREIGN KEY (room_id) REFERENCES tb_room(room_id)
    ON UPDATE CASCADE
    ON DELETE SET NULL
```

```
) ENGINE=InnoDB;
```

```
CREATE TABLE tb_payment (  
    payment_id INT AUTO_INCREMENT,  
    resident_id INT NOT NULL,  
    payment_date DATE NOT NULL,  
    payment_amount INT NOT NULL,  
    month_of_rent VARCHAR(30) NOT NULL,  
    payment_status ENUM('Pending', 'Paid') DEFAULT 'Pending',  
    PRIMARY KEY (payment_id),  
    FOREIGN KEY (resident_id) REFERENCES tb_resident(resident_id)  
        ON UPDATE CASCADE  
        ON DELETE CASCADE  
) ENGINE=InnoDB;
```

```
CREATE TABLE tb_complaint (  
    complaint_id INT AUTO_INCREMENT,  
    resident_id INT NOT NULL,  
    complaint_date DATE NOT NULL,  
    complaint_desc TEXT NOT NULL,  
    complaint_status ENUM('In-Progress', 'Completed') DEFAULT 'In-Progress',  
    PRIMARY KEY (complaint_id),  
    FOREIGN KEY (resident_id) REFERENCES tb_resident(resident_id)  
        ON UPDATE CASCADE  
        ON DELETE CASCADE  
) ENGINE=InnoDB;
```

8. SQL Operations (DML SQL)

SQL

```
-- INSERT DATA
```

```
INSERT INTO tb_room (room_number, room_type, monthly_price, room_status) VALUES
```

```
('A-01', 'Exclusive (AC + Private Bathroom)', 1500000, 'Occupied'),  
( 'A-02', 'Exclusive (AC + Private Bathroom)', 1500000, 'Available'),  
( 'B-01', 'Standard (Fan + Shared Bathroom)', 800000, 'Occupied');
```

```
INSERT INTO tb_resident (room_id, resident_name, phone_number, email, origin_address) VALUES  
(1, 'Alya Nurin', '081234567890', 'alya.nurin@email.com', 'Pasuruan, East Java'),  
(3, 'Farhan Wijaya', '085712345678', 'farhan.w@email.com', 'Surabaya, East Java');
```

```
INSERT INTO tb_payment (resident_id, payment_date, payment_amount, month_of_rent,  
payment_status) VALUES  
(1, '2026-06-01', 1500000, 'June 2026', 'Paid'),  
(2, '2026-06-02', 800000, 'June 2026', 'Pending');
```

```
INSERT INTO tb_complaint (resident_id, complaint_date, complaint_desc, complaint_status)  
VALUES  
(1, '2026-06-05', 'Air conditioner (AC) is leaking water.', 'In-Progress');
```

```
-- UPDATE DATA
```

```
UPDATE tb_complaint SET complaint_status = 'Completed' WHERE complaint_id = 1;
```

```
UPDATE tb_payment SET payment_status = 'Paid' WHERE payment_id = 2;
```

```
-- DELETE DATA
```

```
DELETE FROM tb_complaint WHERE complaint_id = 1;
```

9. CRUD Application Code Links

A. Python (CLI Application)

The Python code is located in python folder (<https://github.com/alyanurin60-hash/Boarding-House-Management-Information-System/tree/main/python>)

B. PHP (Web-Based Native Application)

The code is located in php-app folder (<https://github.com/alyanurin60-hash/Boarding-House-Management-Information-System/tree/main/php-app>)